

An Abstract Approach to Dissipation

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Abstract

We suggest that a very natural mathematical framework for the study of dissipation—in the sense of Willems, Moylan and Hill, and others—is that of indefinite quasimetric spaces. Several basic facts about dissipative systems are seen to be simple consequences of the properties of such spaces. Quasimetric spaces provide also one natural context for optimal control problems, and even for “gap” formulations of robustness.

This is very preliminary work, but it seems appropriate to give a conference presentation to elicit comments and suggestions, since we believe that the area has great potential for development.

1 Introduction

In the pioneering work ([10]), Jan Willems introduced an abstract concept of energy dissipation. He did this in order to unify i/o and state space stability, and in particular with the purpose of understanding conceptually the meaning of Kalman-Yakubovich positive-realness (passivity) as well frequency-domain stability theorems in a general nonlinear context. His work was continued by many authors, most notably Hill and Moylan (see e.g. [3, 4]). The notions that arose from this work are now central to nonlinear control, appearing in so-called “nonlinear H^∞ ” theory, the study of “ISS” systems, and modern adaptive control approaches, among many areas. In particular, a very successful workshop at the 1994 CDC dealt with many issues related to dissipative systems.

It is perhaps remarkable that many of the proofs of the basic foundational results, in particular those dealing with the existence of maximal and minimal storage functions, repeatedly perform a certain step. This step involves showing that the triangle inequality holds for a quantity, let’s call it $W(x, z)$, which measures the minimal net change in “energy” needed in order to bring one state x to another state z . This suggests that perhaps what really matters is to have an abstract notion of “oriented distance” $W(x, z)$ and that some of the basic facts can be formulated at that level. Once this is done, the intrinsic features of the problem are more evident, and questions of choice of particular controls become irrelevant. (This is entirely

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analogous to —and includes as a particular case— the definition of a distance in a Riemannian manifold as the infimum of the lengths of paths joining two points, or the similar construction for totally nonholonomic systems, that is, the sub-Riemannian case, in which paths are restricted to be tangent to a distribution. The concept of geodesic is implicit, but the distance itself is an object of interest.) Moreover, the “distance” W gives rise in turn to a topological space: a state z is near a state x if one can go from x to z using only a small amount of energy (or gaining energy). The standard machinery of topology then becomes in principle applicable to the study of dissipative systems.

A complication is that the standard notion of metric space is not appropriate because of the lack of symmetry: the energy needed to go from state x to state z need not be —and in fact almost never will be— the same as needed to go from z to x . (A reverse transition may not even be possible, which one can model as requiring infinite energy, that is, $W(z, x) = +\infty$.) Thus one requires at least the nonsymmetric notion of *quasi*-(pseudo)metric proposed independently by Wilson and Nyemytzki in 1931 and studied by various authors since then, as well as —for studying the associated topologies— the concept of quasi-uniformity proposed by Nachbin in 1946 (see e.g. [6, 11, 7, 1]). Actually, quasimetrics are not sufficient, because the change in energy $W(x, z)$ may be negative as well as positive (energy gain vs. loss). Thus one needs to consider *indefinite* quasimetrics, which we introduce in this note.

Once that one adopts the point of view of quasimetrics, it is clear that most of optimal control would probably also benefit from such an approach. In fact, any nonreversible dynamical process with an associated cost, even problems of graph search, can be so modeled. Many mathematical structures are most naturally amenable to be treated in this way; for instance, the Hausdorff distance between two subsets A and B of a metric space is defined by first defining an oriented distance from A to B and viceversa and then taking the maximum between the two; each of these two oriented measures defines in itself a quasimetric. This has been noticed, though implicitly, in the control literature, among other places in the context of “gap” characterizations of robustness; the *directed gap* between two systems is defined in terms of a Hausdorff-type distance between graphs of behaviors; see e.g. [2, 8], and this quantity characterizes the largest possible change in a plant for which a given controller still stabilizes. The gap metric is then defined as a symmetrization of the directed gap, but clearly the latter is the most natural notion.

More speculatively, the abstract notion permits treating discrete and continuous variables, and even discrete and continuous time (there is no “time” explicitly appearing), and hence might provide one way of generalizing dissipation ideas to the study of “hybrid” systems.

Let me emphasize once more the motivation, before embarking on the abstract nonsense. Given a continuous-time system Σ and a “supply function” $w(x, u)$, dissipativity in the sense of Willems means that there exists a nonnegative real-valued function V of states, called a “storage function,” so that the inequality

$$V(x(t)) - V(x(0)) \leq \int_0^t w(x(s), u(s)) ds \quad (1)$$

holds along all possible trajectories. (There are several technical variants of the concept, for instance depending on whether infinite values are allowed for V or not, but the above inequality is the most important feature.) In problems involving L^2 gains (that is, “ H^∞ ” problems), the state-space is Euclidean space and the supply function w has the form $\gamma^2 \|u\|^2 - \|x\|^2$; if we also assume that $V(0) = 0$ then dissipativity implies that γ is less or equal to the L^2 gain (that

is, operator norm) of the zero-state response of the system. In this manner, the concept of dissipation (and more precisely, the characterization of “extremal” storage functions as solutions of indefinite Hamilton-Jacobi equations) becomes central to the calculation of operator norms and hence to the solution of disturbance attenuation problems. Alternatively, if w involves a product of functions of states and inputs, passive systems result. A good and very current reference for such matters is the new edition of [5].

The starting observation for the present note is that the inequality (1) is equivalent to asking that $V(z) - V(x) \leq W(x, z)$ for all x, z , where¹

$$W(x, z) := \inf \left\{ \int_0^t w(x(s), u(s)) ds \mid u \text{ so that } x \xrightarrow[u]{\sim} z \right\} \quad (2)$$

(the value of $W(x, z)$ is defined as $+\infty$ if z cannot be reached from x) and this W satisfies the triangle inequality, since concatenations of controls can be used to go from x to z if one can go from x to y and from y to z , and the cost is additive. This “optimal cost of a transfer” defines a sort of distance between states. It seems to be better to work directly with W rather than with w : the selection of controls and particular trajectories is not now an issue, and the pure cost structure is abstracted. Going even further, one may wish to not differentiate between the metrics corresponding to two supply functions w if these metrics give rise to the same notion of “nearness” in the sense that the energy required for a state transfer is “small” for one W_1 if and only if it is also “small” for another W_2 ; one is thus led to studying the topologies induced by these metrics. This paper is a preliminary presentation of some ideas in these directions.

For technical reasons, the assumptions of the abstract theory will rule out the case in which there are self-loops with negative cost, i.e. states x for which $W(x, x) < 0$. This would be equivalent, by the triangle inequality, to infinite negative cost $W(x, x) = -\infty$. This assumption does not seem to be too restrictive vis a vis the cases considered in the dissipation literature (and of course in standard optimal control problems, for which $W(x, z)$ is always nonnegative, even for $x \neq z$). It is possible to generalize the setup to allow $W(x, x) = -\infty$, but doing so gives rise to a less elegant framework in which one needs to ensure at each stage that a $+\infty - \infty$ cancellation never occurs.

2 IQMS

Definition 2.1 An IQM on a set X is a function

$$W : X \times X \rightarrow \mathbb{R} \cup \{+\infty\}$$

which satisfies the following properties:

- $W(x, x) = 0$ for all x .
- For all x, y, z ,

$$W(x, z) \leq W(x, y) + W(y, z).$$

An IQM **space** is a pair (X, W) consisting of set X and an IQM W on X . □

¹Here, as in [9], I am using the notation $x \xrightarrow[u]{\sim} z$ to mean that the control u steers x to z , and t is the length of u . I am not including detailed technical assumptions, because they are irrelevant to the abstract framework to be developed.

When the values of W are nonnegative real numbers, an IQM is an object that has been studied extensively, and is usually called (cf. [1]) a *quasi-pseudo-metric space*. (The terminology varies somewhat among authors. The book [7] calls them just quasimetrics, but in the original paper [11] as well as in [1] this latter term is reserved for spaces for which, in addition, the axiom “ $W(x, y) > 0$ for all $x \neq y$ ” is imposed.) Since negative (as well as positive infinity) values are allowed, following the convention of indefinite metric spaces, it seems natural, though extremely cumbersome, to call these objects “indefinite quasi (pseudo)metrics”, and hence the above terminology².

By abuse of terminology, one often says just “the IQM space X ” instead of “ (X, W) ” when W is clear from the context. It is perhaps useful to call $W(x, z)$ the *gain from x to z* and its negative $-W(x, z)$ the *loss from x to z* . One may think of W as a gain or loss of energy, especially in the context to follow. Metric spaces result when W takes only nonnegative values, is symmetric, and $W(x, z) \neq 0$ for all $x \neq z$. None of these properties is required here.

Given (X, W) , it is convenient (and consistent with the intended control applications) to define the **reachability relation** Reach on the set X as the binary relation given by:

$$x\text{Reach}y \iff W(x, y) < \infty .$$

(We read “ $x\text{Reach}y$ ” as “ x can be steered to y ” (or controlled to y , or y is reachable from x .)

2.1 Subintegrability

Fix an IQM space (X, W) . By definition of IQM, for each fixed element $y \in X$, the function $V = W_y : X \rightarrow \mathbb{R} \cup \{+\infty\}$ given by $V(x) := W(y, x)$ satisfies $V(y) = 0$ and $V(z) \leq V(x) + W(x, z)$ for all $x, z \in X$. The concepts associated to dissipativity in control systems arise from asking when such a function is nonnegative.

Definition 2.2 An $x_0 \in X$ is an **initial point** for (X, W) if

$$W(x_0, z) \geq 0 \text{ for all } z \in X .$$

The (possibly empty) set of initial points is denoted by $\mathcal{I}(X, W)$ (or just $\mathcal{I}$ if (X, W) is clear from the context). $\square$

Definition 2.3 Let $S \subseteq X$ be any subset. The IQM space (X, W) is **subintegrable with respect to S** if $S \subseteq \mathcal{I}$. $\square$

Definition 2.4 A function

$$V : X \rightarrow [0, +\infty]$$

so that

$$V(z) \leq V(x) + W(x, z) \text{ for all } x, z \in X . \quad (3)$$

is called a **subpotential** (or a **storage function**, or a **subintegral**) for W . If $S \subseteq X$ is any subset, and if V is a subpotential which also satisfies

$$V|_S \equiv 0 \quad (4)$$

then V is called a **subpotential (for W) with respect to S** . $\square$

²A better name would be desirable!

Note that the subpotentials for (X, W) form a convex set, as is clear from Equation (3).

Remark 2.5 The above terms are motivated by the standard concepts of integrability and potentials. The more standard terminology in control theory is “dissipativity,” typically with respect to the set S consisting of a single “zero energy state” which is the origin in an Euclidean space. Note that in the case of quasi-pseudo-metric spaces ($W \geq 0$), every point is initial, and the concept is meaningless. (And if $W \geq 0$ then $V \equiv 0$ is a subpotential with respect to all subsets.) The function $V \equiv +\infty$ is always a subpotential, but not with respect to any nonempty set; the interest is in applications in which there are points of “least energy” —so that energy can at best increase from initial points.

In a graph or game-theoretic context, it is probably a good idea to call such a $V(x)$ a “guaranteed gain at x .” In these terms, (3) can be read as saying that if one starts with a guarantee $V(x)$ at x and then gains an additional $W(x, z)$, $V(z)$ is a –conservative– guarantee at z . $\square$

Our first objective is to relate subintegrability with respect to a given subset S to the existence of subpotentials with respect to the same S , and more generally to classify the possible such subpotentials. The main elementary results in that regard, see e.g. [10, 4, 3], can be proved using the formalism introduced here.

2.2 Characterizations

For each two nonempty subsets $A, B \subseteq X$, we let

$$W(A, B) := \inf\{W(a, b), a \in A, b \in B\} \in \mathbb{R} \cup \{\pm\infty\}.$$

This is a generalization of the minimum distance between two sets in metric spaces. If the set A consists of only one element, x , we write simply $W(x, B)$, or $W^B(x)$ if B is fixed and we are interested in the “oriented distance” from the point x to B as a function of $x \in X$. Similarly, if $B = \{x\}$, we write $W(A, x)$ or $W_A(x)$.

If A, B are any two sets and z is any element of X for which it does not hold that $W(A, z) = -W(z, B) = \pm\infty$, then the triangle inequality implies that

$$W(A, B) \leq W(A, z) + W(z, B). \quad (5)$$

If A, B, z are such that $W(A, B) < \infty$ and $W(A, z) > -\infty$, then

$$-W(z, B) \leq W(A, z) - W(A, B). \quad (6)$$

Indeed, if either of $W(A, z)$ or $W(z, B)$ equals $+\infty$ or if $W(A, B) = -\infty$ the equality (6) is trivial. Since $W(A, B) < \infty$ and $W(A, z) > -\infty$, we may assume then that both $W(A, z)$ and $W(A, B)$ are finite. In that case (5) can be applied and it shows that $W(z, B)$ cannot be $-\infty$. Thus we may assume that all of the terms are finite, and hence (6) follows from (5).

We specialize Equation (6) to the case when A is a one-element set $\{x\}$ and $B = X$. Observe that $W(A, B) = W(x, X) \leq W(x, x) = 0$, and $W(x, z) \in \mathbb{R} \cup \{+\infty\}$, so (6) indeed applies with these choices. Using the notation

$$\boxed{\underline{V} := -W^X}$$

we conclude that $\underline{V}$ is a subpotential for W .

Remark 2.6 Note that in the case of nonnegative W , the facts that $W(x, z) \geq 0$ for all z and $W(x, x) = 0$ imply $\underline{V} \equiv 0$. This happens in particular for standard optimal control problems, and for quasi-pseudo-metric spaces. $\square$

The subpotential $\underline{V}$ is minimal, that is,

$$\underline{V} \leq V \quad (7)$$

for every subpotential V . Proof: The inequality is trivial if $V(x) = +\infty$, so pick any x for which $V(x) \in [0, +\infty)$. Applying Equation (3),

$$0 \leq V(x) + W(x, z),$$

so taking infimum over z gives $0 \leq V(x) + W^X(x)$, as desired.

Example 2.7 Consider $X = \mathbb{R}$ with $W(x, y) := y - x$. In this case, for each x , $W(x, y)$ is not bounded below, so $\underline{V} \equiv +\infty$. Thus all possible subpotentials are infinite in this example. In terms of the subset $\mathbb{R}_{\geq 0}$ and the same W , one can say that a finite-valued subpotential is the same as a non-expansive map from an IQM space into this space (and in the case of metric spaces, also the same as a nonexpansive map into $\mathbb{R}_{\geq 0}$ with the usual distance). $\square$

In general, for all x , $W^X(x) = -\underline{V}(x) \leq 0$. By definition, x_0 initial means that $W^X(x_0) = 0$; that is,

$$x_0 \in \mathcal{I} \iff \underline{V}(x_0) = 0. \quad (8)$$

So $\underline{V}|_{\mathcal{I}} = 0$, and thus $\underline{V}$ is a subpotential with respect to S whenever (X, W) is subintegrable with respect to S , and conversely, (X, W) is subintegrable with respect to any subset of the zero set of $\underline{V}$.

Assume now that a subset $S \subseteq X$ has been fixed, and let

$$\boxed{\overline{V} := W_S}.$$

Clearly,

$$\overline{V}|_S \leq 0. \quad (9)$$

Note also that $\overline{V} \geq 0$ (and, in particular, $\overline{V}|_S \equiv 0$) if and only if $W(x_0, X) \geq 0$ for all $x_0 \in S$, that is, if $S \subseteq \mathcal{I}$. If $\overline{V} \geq 0$ then, with $A = S$ and $B = \{x\}$, Equation (5) applies (a cancellation $\pm\infty$ cannot happen), and thus Equation (3) holds for $\overline{V}$. Hence, (X, W) is subintegrable with respect to S if and only if $\overline{V}$ is a subpotential with respect to S .

If V satisfies Equation (3) and Equation (4), then, for the same S , $V(z) \leq V(x) + W(x, z) = 0 + W(x, z)$ whenever $x \in S$, from which it follows, by taking infimum over x , that

$$V \leq \overline{V}. \quad (10)$$

Combining Equations (10) and (7), it follows that every possible subpotential with respect to S must satisfy

$$\boxed{\underline{V} \leq V \leq \overline{V}}. \quad (11)$$

The next remark summarizes the main abstract properties of subintegrability.

Theorem 1 Fix an IQM space (X, W) , and consider a subset $S \subseteq X$. The following properties are equivalent:

1. W is subintegrable with respect to S .
2. $W(x_0, x) \geq 0$ for all $x_0 \in S$ and $x \in X$.
3. $\overline{V} \geq 0$.
4. $\underline{V}|_S \equiv 0$.
5. $\overline{V}$ is a subpotential with respect to S .
6. $\underline{V}$ is a subpotential with respect to S .
7. There is some subpotential with respect to S .
8. $\underline{V} \leq \overline{V}$.

Proof. Equation (8) shows that 1 is equivalent to 4. As remarked above, 1, 2, and 3 are equivalent, immediately from the definition of $\overline{V}$, and in that case 5 holds. It was also remarked that 1 implies 6. Obviously, 5 and 6 each imply 7, which in turn, because of Equation (11), implies 8. The proof is completed by noticing that since $\underline{V}$ is always nonnegative, 8 implies 3. ■

It is natural to ask when there is any finite subpotential.

Proposition 2.8 The following statements are equivalent:

1. There is a finite-valued subpotential.
2. $\underline{V} : X \rightarrow [0, +\infty)$.
3. W has **finite-loss**, i.e., for each $x \in X$ the set $\{W(x, z), z \in X\}$ is bounded below.

Proof. Clearly, (2) implies (1), and since $W^X(x)$ is the infimum of the values $\{W(x, z), z \in X\}$, (3) implies (2). On the other hand, $W(x, z) \geq V(z) - V(x) \geq -V(x)$ shows that (1) implies (3). ■

Remark 2.9 Regarding the definition of $\underline{V}$, observe that, in general, one can restrict the infimum to: $-\underline{V}(x) = \inf_{z \in R(x)} W(x, z)$, where $R(x)$ is the *reachable set from x* , i.e. the set of z so that $W(x, z) < \infty$ (this set is always nonempty since $x \in R(x)$). Take any x for which $\underline{V}(x) < \infty$ and any $z \in R(x)$; Equation (3) applied to $V = \underline{V}$ then gives

$$0 \leq \inf_{z \in R(x)} \underline{V}(z) \leq \underline{V}(x) + \inf_{z \in R(x)} W(x, z) = 0$$

so

$$\inf_{z \in R(x)} \underline{V}(z) = 0$$

for each such x . □

Remark 2.10 When there is “reachability from S ” i.e., $\overline{V}$ is finite ($W(x_0, x) < \infty$ for all $x_0 \in S$ and all $x \in X$), all subpotential functions are automatically finite (because of Equation (11)). In general, subpotential functions are finite on the reachable set $R(S)$. $\square$

Remark 2.11 Denote $C(y) := \{x \mid W(x, y) < \infty\}$, the set of points “controllable to” y . Then, for each $y, z \in X$,

$$\max_{x \in C(y)} [W_x(z) - W_x(y)] = W(y, z) . \quad (12)$$

In general, from the triangle inequality, $W(x, z) \leq W(x, y) + W(y, z)$. If $W(x, y) < \infty$ one has also that $W_x(z) - W_x(y) \leq W(y, z)$, and in particular, for $y \in C(y)$, equality holds. Thus (12) is true. $\square$

2.3 Integrable Case

Definition 2.12 A function

$$V : X \rightarrow [0, +\infty)$$

which satisfies

$$V(z) = V(x) + W(x, z) \quad \text{for all } x, z \in X \quad (13)$$

is called a **potential** (or an **integral**) for W . $\square$

This is perhaps a less interesting concept in control applications, but it helps in understanding subpotentials. A potential for W is in particular a subpotential.

The IQM W is **independent of path** if

$$W(x, z) = W(x, y) + W(y, z) \quad \text{for all } x, y, z .$$

It is **antisymmetric** if

$$W(x, z) = -W(z, x) \quad \text{for all } x, z$$

(this implies that W can attain only finite values).

Proposition 2.13 The following properties are equivalent:

1. There is a potential V for W .
2. W is independent of path and finite-loss.
3. W is antisymmetric and finite-loss.

Proof. If there is a potential V for W , then W is independent of path, as follows from

$$V(z) - V(x) = [V(y) - V(x)] + [V(z) - V(y)] .$$

Also, in that case W admits a finite-valued subpotential, so it is finite-loss. Independence of path implies antisymmetry, since

$$W(x, z) + W(z, x) = W(x, x) = 0$$

for each x, z .

Finally, assume that W is antisymmetric and finite-loss. Pick any finite-valued subpotential. This must be a potential. Indeed, by definition of subpotential, $V(z) - V(x) \leq W(x, z)$; on the other hand, $V(x) - V(z) \leq W(z, x) = -W(x, z)$. Thus $V(z) - V(x) = W(x, z)$. $\blacksquare$

Remark 2.14 In general, if $V : X \rightarrow \mathbb{R}$ is bounded below then $W(x, z) := V(z) - V(x)$ is independent of path and finite-loss. This holds in particular when $V = W_y$, for any $y \in X$, if W is any IQM which is finite-valued and finite-loss (equivalently, W admits a finite-valued subpotential, cf. Proposition (2.8)), thus inducing a W as in Proposition (2.13) via the formula

$$W_{[y]}(x, z) = W(y, z) - W(y, x) .$$

By Equation (12), for each x, z it holds that

$$W(x, z) = \max_{y \in X} W_{[y]}(x, z) .$$

The conclusion is that if W is finite-valued and finite-loss then it has a representation of the form $W = \max_{\lambda \in \Lambda} W_\lambda$ in terms of IQMs which satisfy the integrability conditions in Proposition (2.13). $\square$

One can often normalize the potential:

Lemma 2.15 If there is a potential V for W which attains its minimum at a set S , then W is subintegrable with respect to S . Conversely, if W is subintegrable with respect to S , and if there is a potential for W , then there is a potential V for W so that $V|_S \equiv 0$.

Proof. Assume that V is a potential and α is the minimum value of V . Pick any $x_0 \in S = V^{-1}(\alpha)$. Then, $W(x_0, z) = V(z) - V(x_0) \geq 0$ for all z , so $x_0 \in \mathcal{I}$. Conversely, assume that W is subintegrable with respect to S and let V be any potential. For each $x_0 \in S$, and any x , $V(x) = V(x_0) + W(x_0, x) \geq V(x_0)$, so V is constant and attains its global minimum value, say α , on S . The function $V'(x) := V(x) - \alpha$ is again a potential for W , and it holds that $V'(S) = 0$. $\blacksquare$

Uniqueness of V

Assume that W is subintegrable with respect to a given subset S . It is interesting to ask when is there a *unique* possible subpotential with respect to S . (Because of Equation (11), this is equivalent to asking when is $\overline{V} = \underline{V}$.)

Assume that W is known to admit a potential V and to be subintegrable with respect to S . Then $\overline{V} = \underline{V}$. To show this, first apply Lemma (2.15) to normalize V and so assume $V|_S \equiv 0$. Pick any $x_0 \in S$ and any $x \in X$. Then

$$W(x_0, x) = V(x) - V(x_0) = V(x) .$$

Thus $\overline{V}(x) = \inf_{x_0 \in S} W(x_0, x) = V(x)$ for all x . It follows that every V is equal to $\overline{V}$.

A simple –but interesting from a control-theory viewpoint– necessary condition can be stated as well. Consider the following property:

$$\text{for all } x \in X, \quad W(S, R(x)) = 0 . \tag{14}$$

Equivalently: $\inf_{z \in R(x)} \overline{V}(z) = 0$ for all x . One can interpret this as an “asymptotic controllability to S ” property. It says that from each point x one can get as “close” as desired to the set S (in applications, S is some desired set of equilibria for a dynamical system; “close” can be interpreted in terms of one of the topologies defined later). If it is the case that $\overline{V} = \underline{V} < \infty$ then from Remark (2.9), property (14) holds.

To summarize, we have that antisymmetry implies uniqueness of subpotential, which in turn implies asymptotic controllability to S .

2.4 Brief Remarks for Control Systems

Some informal remarks can be made about the case of control systems, for which W is defined as in Equation (2), given an underlying continuous time system $\dot{x} = f(x, u)$.

Remark 2.16 Consider systems without drift, that is, with equations $\dot{x} = \sum_i u_i g_i(x)$. For such systems, $x \xrightarrow{u} z$ is an equivalence relation (the reverse transition can be achieved using the time-reversed negative of u). Assume that the supply function w depends only on u and is odd. Under suitable assumptions on w plus controllability (the totally nonholonomic case in which the Lie algebra generated by the g_i 's has full rank at each point), W is finite and antisymmetric. Note on the other hand that typical optimal control problems can never be antisymmetric in this sense, since for such, W is nonnegative. $\square$

Remark 2.17 With $S = \{0\}$, asymptotic null controllability in the above sense happens if 0 is in the closure of the reachable set and $\bar{V}$ is continuous at zero. Also note that in that case also $\inf_{z \in R(x)} V(z) = 0$ for all subpotential functions V . $\square$

Remark 2.18 In general W is indefinite. Assume that w is a gain-type supply $w(x, u) = \gamma^2 |u|^2 - |x|^2$. (It is assumed that the states space is an Euclidean space and controls are given an L^2 norm.) Then “ W is subintegrable” with respect to the set $S = \{0\}$ is the same as saying that γ is less or equal to the L^2 gain of zero-state response of the system. This is clear from the fact that the inequality $\bar{V}(x) = W(0, x) \geq 0$ is the same as asking that

$$\|x_T\|_2^2 = \int_0^T |x(s)|^2 ds \leq \gamma^2 \int_0^T |u(s)|^2 ds = \gamma^2 \|u_T\|_2^2$$

for all controls. (Subscript T indicates restriction to interval $[0, T]$.) Note that, if γ is exactly equal to the gain, then $\bar{V}$ cannot be positive definite. Indeed, pick any (nonzero) control u achieving the gain, i.e. $\|x_T\|_2 = \gamma \|u_T\|_2$; it follows that along this particular trajectory the integral in the inf in (2) is zero, so also $W(0, x(T)) = 0$, i.e. $\bar{V}(x) = W(0, x)$ is not positive for all x . $\square$

Remark 2.19 Classically one uses dissipation arguments in order to obtain asymptotic stability results for the unforced system $\dot{x} = f(x, 0)$ (cf. [10, 4, 3]). Let $\phi(t, x)$ be the solution of this system with initial value x , assuming this is defined for all $t \geq 0$. Thus one considers

$$W_0(x, \phi(t, x)) := \int_0^t w(x(s), 0) ds \tag{15}$$

along solutions. Note that

$$W_0(x, \phi(t, x)) \geq W(x, \phi(t, x)). \tag{16}$$

Axiomatically, assume given for each x a curve $\phi(t, x)$, and a function W_0 so that (16) holds whenever $\phi(t, x)$ is defined. Then, if $W_0 \leq 0$ (which is the case for instance for gain-type supply functions w), it follows that any subpotential V is nonincreasing along trajectories, $V(\phi(t, x)) \leq V(x)$ and is hence a candidate Lyapunov function. If in addition V is continuous and $\phi(t, x) \rightarrow 0$ as $t \rightarrow +\infty$, also V is nonnegative. Finally, if there is a detectability condition (i.e., $W_0(x, \phi(t, x)) \equiv 0$ implies $\phi(t, x) \rightarrow 0$ as $t \rightarrow +\infty$) then asymptotic stability results via an invariance principle argument. $\square$

Remark 2.20 If X is also a metric space with distance function d and

$$\text{dist}(x, z) \text{ small} \Rightarrow W(x, z) \text{ small}$$

(this is a “strong local controllability” condition) then *every subpotential is continuous*. This is clear because in that case both $V(x) - V(z) \leq W(x, z)$ and $V(z) - V(x) \leq W(z, x)$ are small for x near z . $\square$

3 More on IQMs

Given the interest of the notion, it seems appropriate to explore further the concept of IQM space from a more purely-mathematical standpoint.

3.1 New IQMs from old ones

In general, if $\alpha : \mathbb{R} \cup \{+\infty\} \times \mathbb{R} \cup \{+\infty\} \rightarrow \mathbb{R} \cup \{+\infty\}$ has $\alpha(0, 0) = 0$ and is monotone:

$$a \leq c \text{ and } b \leq d \Rightarrow \alpha(a, b) \leq \alpha(c, d)$$

and subadditive:

$$\alpha(a + b, c + d) \leq \alpha(a, c) + \alpha(b, d)$$

then for each two IQMs W_1 and W_2 on X ,

$$\alpha(W_1, W_2)(x, y) := \alpha(W_1(x, y), W_2(x, y))$$

is again an IQM. This construction can be applied to $\alpha(a, b) = \rho a + b$, with $\rho > 0$, showing that IQMs form a convex cone.

It also follows that $\max\{W_1, W_2\}$ is an IQM for each two IQMs W_1 and W_2 . In particular, since $W \equiv 0$ is itself an IQM, one can define the **positive part** of an IQM W by

$$W_+ := \max\{W, 0\}.$$

For each *nonnegative* IQM W , $\min\{W, 1\}$ is also an IQM, and in particular

$$W_+^1 := \min\{W_+, 1\}$$

is a quasi-pseudo-metric in the sense of [1] (or just a quasi-metric in the terminology used in [7]).

Defining

$$W^\dagger(x, y) := W(y, x)$$

makes $(X, W^\dagger)$ into another IQM space. the **transpose** of (X, W) . Taking

$$\widetilde{W} := \max\{W, W^\dagger\}$$

provides the **symmetrization** of (X, W) . Since

$$0 \leq W(x, x) \leq W(x, y) + W(y, x), \tag{17}$$

it follows that $\widetilde{W}(x, y) \geq 0$ for all x, y . Furthermore, $\widetilde{W}$ is obviously symmetric, so $\widetilde{W}$ is a pseudometric in the usual sense (but with possible infinite values), and $\min\{\widetilde{W}, 1\}$ is a pseudometric.

A rich source of examples is provided by the following construction. Let (X, W) be any IQM space and assume that R is a transitive and reflexive relation on X . Define

$$W_R(x, y) := \begin{cases} W(x, y) & \text{if } xRy, \\ +\infty & \text{otherwise.} \end{cases}$$

This gives a new IQM space $X_R = (X, W_R)$. Every space arises in this form, since $X_{\text{Reach}} = X$.

Note that $(W_R)^\dagger = W_{R^\dagger}^\dagger$, where $xR^\dagger y$ means that yRx , and $\widetilde{W}_R = \widetilde{W}_{\widetilde{R}}$, where $\widetilde{R} = R \cap R^\dagger$.

For any (X, W) , denote

$$x \sim y \iff W(x, y) = W(y, x) = 0.$$

This is a reflexive and symmetric relation. Furthermore, because of (17), $x \sim y$ iff both $W(x, y) \leq 0$ and $W(y, x) \leq 0$; thus, by the triangle inequality it follows that $\sim$ is transitive and hence is an equivalence relation. Moreover, $x \sim y$ and $u \sim v$ implies $W(x, u) = W(y, v)$. Thus there is a well-defined induced IQM in $X/\sim$, when one defines $(W/\sim)([x], [y]) := W(x, y)$.

3.2 Topologies

Unless otherwise stated, from now on an IQM space (X, W) is fixed. Often one is interested in those properties of W which remain invariant under small perturbations (changes in the supply rate w , for instance). A topology can be introduced in order to deal with such questions, as well as, more generally, to understand the meaning of “closeness” in the W metric.

Consider the following “ball of radius ε ”:

$$B_\varepsilon(x) := \{y \mid W(x, y) < \varepsilon\}$$

for each $x \in X$ and each $\varepsilon \in \mathbb{R}$. These sets form a basis for a topology, since they cover the space ($x \in B_\varepsilon(x)$ whenever $\varepsilon > 0$) and intersections are unions of such balls (for any $x, y \in X$ and $\mu, \nu \in \mathbb{R}$, and for each $z \in B_\mu(x) \cap B_\nu(y)$, let $\varepsilon = \min\{\mu - W(x, z), \nu - W(y, z)\} > 0$; then it holds that $B_\varepsilon(z) \subseteq B_\mu(x) \cap B_\nu(y)$). We’ll call this the **(right) topology induced by (X, W)** , and by abuse of terminology talk about the “topology of X .”

The above remarks show also that using only the balls with $\varepsilon > 0$, the same topology is generated. This implies in particular that the topology induced by W is the same as the topology induced by W_+ . Moreover, balls of small positive radius already form a basis for the topology, so W_+^1 also induces the same topology as W . We conclude that the topological spaces induced by IQMs are precisely the same as what are called *quasi-pseudo-metrizable spaces* in the literature.

The quasimetrization problem is that of determining necessary and sufficient conditions in order that a topological space admit a quasi-pseudo-metric. There is at present no completely satisfactory solution of this question, but a considerable body of knowledge exists. Briefly, a *quasiuniformity* on a set X is a family of subsets $\mathcal{U}$ of $X \times X$ closed under intersections, containing the diagonal, and with the properties that: (a) $U \in \mathcal{U}$ and $U \subseteq V \Rightarrow V \in \mathcal{U}$, and (b)

for each $U \in \mathcal{U}$ there is some $V \in \mathcal{U}$ so that $V \circ V \subseteq U$ (where the composition is understood in the sense of binary relations on the set X). Given a quasi-pseudo-metric W , the smallest quasiuniformity $\mathcal{U}$ containing all the sets $\{(x, y) \mid W(x, y) < \varepsilon\}$ is the quasiuniformity induced by W . A topology is associated to each quasiuniformity in the obvious manner, namely by taking as neighborhoods of a point x those sets of the form $U(x) = \{y \mid (x, y) \in U\}$, for each $U \in \mathcal{U}$. This coincides with the above topology, if the quasiuniformity is induced by W . An analogue of A. Weil's theorem on uniformization is valid, in stronger form: a quasiuniformity $\mathcal{U}$ admits a quasi-pseudo-metric if and only if $\mathcal{U}$ admits a countable base; see [7], Theorem 11.1.1. (It turns out that every topological space arises from some quasiuniformity, see [7], Theorem 11.1.2, but the base in this construction is in general not countable.)

The quotient $X/\sim$ is always a “Kolmogoroff space” in the sense of Bourbaki (also called an “Axiom T_0 space”): for each $x \neq y$ in the quotient, either $W(x, y) > 0$ or $W(y, x) > 0$, so either there is a neighborhood of x not containing y or viceversa. However, neither X nor $X/\sim$ need be Hausdorff, nor even Axiom T_1 (“points are closed sets”); the latter property is equivalent to asking that there does not exist any pair of distinct elements x, y with $W(x, y) \leq 0$. Observe that (X, W) is of class T_1 iff (X, W_+^1) is, which is in turn equivalent to asking that W_+^1 be a quasimetric (as opposed to a quasi-pseudo-metric); the book [1], chapter 7, studies in detail characterizations of quasiuniformities induced by quasimetrics.

Other Associated Topologies, and some Examples

Dually, one may define the balls

$$B^\varepsilon(x) := \{y \mid W(y, x) < \varepsilon\}$$

and a corresponding **left topology**, which is the same as the right topology for the transpose $(X, W^\dagger)$. All statements for right topologies will admit obvious dual statements by means of transposition, so we mostly use the right topology. One can also define the **bilateral topology** as the smallest topology that contains both the right and left topologies. This is the same as the topology induced by the pseudometric $\widetilde{W}$, and a basis for it is obtained by considering all bilateral balls

$$B_\varepsilon(x) \cap B^\varepsilon(x) = \{y \mid W(x, y) < \varepsilon \text{ and } W(y, x) < \varepsilon\}.$$

We may also consider the balls defined by nonstrict inequalities: $\overline{B}_\varepsilon(x) := \{y \mid W(x, y) \leq \varepsilon\}$, and $\overline{B}^\varepsilon(x) := \{y \mid W(y, x) \leq \varepsilon\}$. There is no reason for $\overline{B}_\varepsilon(x)$ to be closed for the right topology. However, for each $\varepsilon \in \mathbb{R}$ and $x \in X$, $\overline{B}^\varepsilon(x)$ is closed, as proved below for the more general case of distance to sets. Transposing, also $\overline{B}_\varepsilon(x)$ is closed for the *left* topology.

Some examples can be useful in understanding the generality of these concepts.

Example 3.1 Given any set X and function $H : X \rightarrow \mathbb{R}$, $W(x, y) := H(y) - H(x)$ defines a IQM on X . Now assume that R a transitive and reflexive relation on X . Then W_R corresponds to $W_R(x, y) = H(y) - H(x)$ if xRy and $:= +\infty$ otherwise. Observe that the transposed space is obtained in the same way by considering $R^\dagger$ and $-H$ instead of H . Furthermore, $\widetilde{W}_R(x, y) = |H(y) - H(x)|$ if $x\widetilde{R}y$ and $\widetilde{W}(x, y) = +\infty$ otherwise. $\square$

Example 3.2 In example (3.1), take $X = \mathbb{R}$, $H(x) = x$, and the relation R defined by $xRy \iff x \leq y$. The right topology has as a basis the intervals of the form $[x, y)$, the

left topology the intervals $(y, x]$, and the bilateral topology is the discrete topology. All the three topologies are Hausdorff. Convergence “ $x_n \rightarrow x$ ” in the right topology means that $x_n \rightarrow x$ in the usual sense and $x_n \geq x$ for almost all n . The right topology is in this case the same as the topology called $\mathcal{T}_+$ in Exercise 5 in Bourbaki, Topology, I.2, for general ordered sets, and specifically treated for the case of the reals in Exercise 16 of IX.5 (to be precise, the latter exercise deals with the corresponding left topology). It is remarked there that open sets are disjoint unions of intervals open on the right, and the topology is “perfectly normal” (hence regular) and totally nonmeagre (hence Baire), separable, Lindelöf, and paracompact, but it is not metrizable. $\square$

Example 3.3 Again in example (3.1) take $X = \mathbb{R}$ and $H(x) = x$, but now let xRy for all x, y . Now the right topology has as a basis the intervals of the form $(-\infty, x)$, dually for the left topology, and the the bilateral topology coincides with the usual topology in $\mathbb{R}$. The one-sided topologies are not Hausdorff. [Note: this is not the same as the “order topology” defined in Exercise 2 in Bourbaki, Topology, I.1 for linearly ordered sets, using as basis the closed intervals $(-\infty, x]$.] In this example, there are no nonconstant continuous maps $\phi : X \rightarrow \mathbb{R}$ for the right topology (the same conclusion holds for the left topology). Indeed, assume that ϕ would be continuous and that there are a, b so that $\phi(b) < \phi(a)$. Since $\phi(b) \in (-\infty, \phi(a))$, b is in $\phi^{-1}(-\infty, \phi(a))$. The open sets are precisely those of the form $(-\infty, c)$, so if it were the case that $a \leq b$ then also $a \in \phi^{-1}(-\infty, \phi(a))$, a contradiction. Thus $a > b$. On the other hand, $a \in (\phi(b), +\infty)$ implies that a is in $\phi^{-1}(\phi(b), +\infty)$, which must also be open and hence of the form $(-\infty, c)$, so $b < a$ implies the contradiction $b \in \phi^{-1}(\phi(b), +\infty)$. So in general there are no continuous functions separating points from disjoint closed sets, which in particular means that the one-sided topologies do not arise from uniformities, and hence are in particular not metrizable. $\square$

Remark 3.4 Another application of the equality (12) is as follows. One may define “quasi-normed spaces” by starting with a real vector space X and postulating a subadditive map $x \mapsto \|x\|$ (such functions arise for instance as gauge or Minkowski functionals for not-necessarily balanced subsets of X). Then $\|x - y\|$ defines an IQM on X . Starting from any finite-valued W , the map

$$y \mapsto \{W_x(y), x \in X\}$$

provides an isometry from (X, W) into a “generalized l^∞ ” quasinormed space. $\square$

Regularity of Distance to a Set

For any $A \subseteq X$ and $\varepsilon \in \mathbb{R}$, let

$$B_\varepsilon(A) := \{y \mid W(A, y) < \varepsilon\} = \bigcup_{x \in A} B_\varepsilon(x)$$

where the last equality, which is clear from the definition, shows that each such set is open (for the right topology). (Similar considerations apply to $B^\varepsilon(A)$, defined in the obvious manner, with respect to the left topology.) Consider the maps

$$W_A : X \rightarrow \mathbb{R} \cup \{\pm\infty\} : x \mapsto W(A, x) .$$

Since $W_A^{-1}(-\infty, \varepsilon) = B_\varepsilon(A)$ is open, each such map is upper semicontinuous.

Consider also the sets, for any $A \subseteq X$ and $\varepsilon \in \mathbb{R}$,

$$\overline{B}^\varepsilon(A) := \{y \mid W(y, A) \leq \varepsilon\}.$$

We claim that such a set is closed (in the right topology). Indeed, its complement is $\{y \mid W(y, A) > \varepsilon\}$, and an element $x \in X$ is in this complement if and only if

$$B_{\varepsilon+\delta}(x) \cap A = \emptyset \text{ for some } \delta > 0.$$

Given such x, δ , $B_{\varepsilon+\delta/2}(B_{\delta/2}(x)) \cap A = \emptyset$, so $B_{\delta/2}(x)$ is an open neighborhood of x with the same property, establishing that the complement is open. Consider the maps

$$W^A : X \rightarrow \mathbb{R} \cup \{\pm\infty\} : x \mapsto W(x, A).$$

Since $(W^A)^{-1}(\varepsilon, +\infty) = \text{complement of } \overline{B}^\varepsilon(A)$ is open, each such map is lower semicontinuous. In particular, if $A \subseteq X$ is quasicompact (i.e., compact but not necessarily Hausdorff) in the right topology, and B is arbitrary,

$$\min_{a \in A} W^B(a) = \inf_{a \in A} W^B(a) = W(A, B).$$

Transposing, each W_A is lower semicontinuous in the left topology, so if $B \subseteq X$ is quasicompact in the left topology,

$$\min_{b \in B} W_A(b) = \inf_{b \in B} W_A(b) = W(A, B)$$

as well. In particular, if $A \subseteq X$ is right-quasicompact and $B \subseteq X$ is left-quasicompact then $W(A, B) = W_A(b_0)$ for some $b_0 \in B$, and applying that W^{b_0} is lower semicontinuous, there must be an $a_0 \in A$ so that $W^{b_0}(a_0) = W(A, b_0)$, which equals $W_A(b_0)$. Thus in this case the minimum is attained:

$$W(A, B) = \min_{a \in A, b \in B} W(a, b).$$

In particular, if $W(a, b) > 0$ for each $a \in A, b \in B$ then $W(A, B) > 0$, provided that A be right-quasicompact and B left-quasicompact. These properties can be used to establish the elementary existence theorems for optimization on IQM spaces.

4 Some Remarks

A theory, in the style of metric geometry, can be in principle developed, in which one defines lengths of paths in terms of the given IQM, then corresponding “geodesic” curves, and even a notion of differentiability and a Hamilton-Jacobi equation for dynamic programming. Obviously, details cannot fit in this conference paper; a longer paper is in preparation.

It seems worth to conclude by providing the following quotation from [11], which is given right after the definition of quasimetrics: *“Usually the result of such a limitation on a set of axioms is to diminish the number of theorems easily deducible, but in this case there is an embarrassing richness of material.”*

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